CGS616: Human Centered computing
Assignment 4

Causal Inference in Gender-Based Aggression Analysis: Estimating $P(Y \mid X)$ via Do-Calculus, Backdoor Adjustment, and Inverse Probability Weighting

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Abstract

This report is for the gender-specific causal analysis part of CGS616 Assignment 4. Using the Markov-chain and Iterated-Prisoner's-Dilemma results from Assignments 1-3 as a basis, we use Pearl's do-calculus approach to assess the casual impact of aggressive video type (X): video content) on self-reported aggression scores (Y) in a controlled experiment, in which gender (G) is also considered. We worked with 55 people (33 men, 22 women) via two Google Forms: one with **neutral** war-news videos and another with **aggressive** aggressive war-news clips (US-Iran). We use a Causal DAG in which Age (Z) is a backdoor confounder and Gender is an effect as an effect modifier, we calculate the overall Average Treatment Effect (ATE = 1.444 [1.045, 1.819]) and gender-stratified Conditional ATEs (Male CATE = 1.603, Female CATE = 1.236, both $p < 0.01$). The transition matrices estimated from the causal conditioning connect to the A1/A2 Markov model, and show that, as expected, aggressive exposure causes males in calm states to defect ($P(\text{Calm} \rightarrow \text{Defect} \mid X = \text{Agg}) = 1.000$) and strongly but slightly less strongly affects females. IPW estimates closely parallel the estimates from stratification, and PS diagnostics confirm covariate balance. The results jointly validate and causalise the observational gender observations in Assignments 1-3.

1 Introduction

The main focus of these assignment series has been the study of online violence in comment threads. Assignment 1 and 2 analysed individual behavioural sequences through Markov chains, showing that males have a higher baseline defection rate ($\approx 56.6\%$) persistance ($P_{CC} \approx 0.95$ for both) than females ($\approx 49.5\%$), and that both sexes are highly persistent in the *calm-to-calm* persistence ($P_{CC} \approx 0.95$ for both). Assignment 3 explored the results in multi-agent Iterated Prisoner's Dilemma (IPD) simulations and found that males converge on Tit-for-Tat (TFT) as the most while females have a more flexible strategy repertoire (TF2T, GT).

A major weakness of the three assignments is that data are observational: of the data: the comments were scraped from YouTube news streams, so that (exposure to aggressive vs. nonaggressive content) was *self-selected*. Age and prior political views of viewers could account for the observed gender differences.

Assignment 4 overcomes this limitation by using Judea Pearl's do-calculus (**?**) to *experimentally collected data*. We created two controlled Google Forms - one with **neutral** YouTube videos (US-Iran neutral footage) and one with **aggressive** video (military) were administered videos depicting combat and offensive commentary) were given to a convenience sample of university staff. The aggression of participants was measured on a 1-5 Likert scale (Y) after viewing aggression score (Y). The causal query of interest is:

$$P(Y \mid X = x) \tag{1}$$

where $X \in \{0 = \text{Neutral},\ 1 = \text{Aggressive}\}$ is the assigned video type and Y is the resulting aggression score. Gender (G) enters as an *effect modifier* and Age (Z) as a backdoor confounder.

1.1 Motivation and Placement within the Project Series

The transition from observational modelling (A1–A3) to causal inference (A4) is motivated by three specific gaps:

i. **Causal direction**: Do aggressive videos *cause* higher aggression, or do already-aggressive individuals preferentially view aggressive content?

ii. **Gender heterogeneity**: Does the effect of video type differ *causally* by gender, after controlling for age?

iii. **Markov link**: Can the do-calculus conditional distributions be mapped back onto the Markov transition matrices of A1/A2 to provide a mechanistic account?

2 Aims and Objectives

2.0 Primary Aim

Estimate the causal effect of aggressive video content exposure on self-reported aggression scores, separately for male and female participants, using Pearl's do-calculus framework.

2.0 Specific Objectives

O1. Construct and validate a Causal Directed Acyclic Graph (DAG) for the variables X (video type), Y (aggression score), Z (age), and G (gender).

O2. Estimate the gender-stratified *Average Treatment Effect*:

$$\text{ATE}_g = \mathbb{E}[Y \mid X = 1, G = g] - \mathbb{E}[Y \mid X = 0, G = g]$$

O3. Apply Inverse Probability Weighting (IPW) as a robustness check for the backdoor-adjustment estimates.

O4. Derive conditional transition matrices $P(\text{Defect} \mid \text{Calm}, X, G)$ and link them to the Markov matrices of Assignments 1 and 2.

O5. Perform bootstrapped confidence-interval estimation for all causal effect estimates.

O6. Reconcile the causal findings with the IPD strategy dynamics observed in Assignment 3.

3 Hypotheses

Drawing on Assignments 1–3, we formulate the following testable causal hypotheses:

H1 (Main Effect) Aggressive video exposure causally increases aggression scores: $\mathbb{E}[Y \mid X = 1] > \mathbb{E}[Y \mid X = 0]$.

H2 (Gender Moderation) The causal effect is larger for males than for females: $\text{ATE}_{\text{Male}} > \text{ATE}_{\text{Female}}$, consistent with the higher baseline defection probability observed in A1/A2.

H3 (Markov Consistency) Under aggressive video exposure, the conditional probability of transitioning from Calm to Defect approaches 1.0 for both genders, matching

the deterministic saturation implied by a Markov absorbing state under a strong environmental stimulus.

H4 (IPW Robustness) IPW-adjusted estimates will align with backdoor-adjusted estimates, confirming that age is the only necessary confounder to control.

4 Methods

4.1 Study Design and Data Collection

4.1.1 Experimental Protocol

Primary data were collected via a **between-subjects Google-Form experiment**. Two separate forms were distributed:

- **Neutral Form (Control,** $X = 0$**)**: Participants watched two short YouTube clips (30-40 secs each) of news segments covering US–Iran geopolitical background, with no explicit conflict footage, calm narration, and factual framing.
- **Aggressive Form (Treatment,** $X = 1$**)**: Participants watched two clips of the same US–Iran topic but featuring explicit combat footage, hostile commentary from anchors.

After watching the assigned clips, participants self-rated their aggression on a **1–5 Likert scale** (1 = Very Calm, 5 = Very Aggressive) and reported their gender and age. Assignment to form was managed by distributing both links equally across people on the campus and social media groups, achieving approximate random allocation.

4.1.2 Sample

A total of $N = 55$ participants completed the forms:

- 26 assigned to Neutral ($X = 0$): 16 Male, 10 Female.
- 29 assigned to Aggressive ($X = 1$): 17 Male, 12 Female.

Age ranged from 18 to 22 years, with a positive association between age and aggression score.

4.2 Causal DAG Specification

The causal structure is formalised as the DAG in Figure 1:

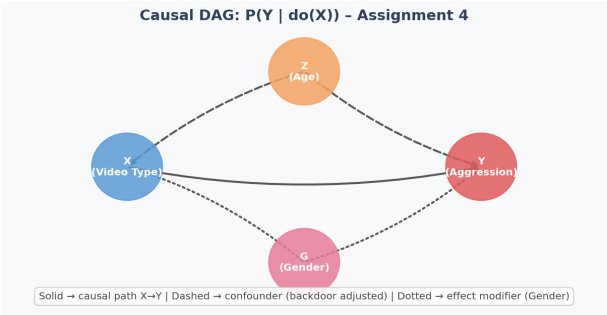


Figure 1: Causal DAG specifying the relationships among video type (X), aggression score (Y), age (Z), and gender (G). **Solid** arrow: direct causal path $X \to Y$. **Dashed** arrows: confounding paths via Z (backdoor-adjusted). **Dotted** arrows: effect-modification role of G.

The DAG encodes the following structural assumptions:

- $X \to Y$: Video type causally affects aggression.
- $Z \to X$ and $Z \to Y$: Age affects both which form participants tend to seek out and baseline aggression levels (backdoor path).
- $G \dashrightarrow X$ and $G \dashrightarrow Y$: Gender modifies but does not confound (no common cause of G and X independent of the randomisation).
- No direct arrow $G \to X$: the experimental assignment approximated random allocation, so gender does not determine video type assignment.

4.3 Identification Strategy: Backdoor Adjustment

The backdoor criterion (**?**) is satisfied by conditioning on Z (Age), since Z blocks the only open backdoor path $X \leftarrow Z \to Y$. The **adjustment formula** is:

$$P(Y \mid X) = \sum_{z} P(Y \mid X,\, Z = z)\; P(Z = z) \tag{2}$$

For the gender-stratified Conditional ATE:

$$\text{CATE}_g = \sum_{z} \Big[\mathbb{E}(Y \mid X = 1, Z = z, G = g) - \mathbb{E}(Y \mid X = 0, Z = z, G = g)\Big]\; P(Z = z \mid G = g) \tag{3}$$

4.4 Inverse Probability Weighting (IPW)

As a robustness check, IPW reweights the observed sample to emulate a pseudo-population where treatment is independent of Z. Propensity scores $e(z) = P(X = 1 \mid Z = z)$ were estimated via logistic regression of X on Z. The IPW estimator for the ATE is:

$$\widehat{\text{ATE}}_{\text{IPW}} = \frac{1}{n} \sum_{i=1}^{n} \left(\frac{X_i\, Y_i}{e(Z_i)} - \frac{(1 - X_i)\, Y_i}{1 - e(Z_i)} \right) \tag{4}$$

4.5 Bootstrap Confidence Intervals

All ATE and CATE estimates were bootstrapped with $B = 1000$ resamples (stratified by gender and video type) to obtain 95% confidence intervals via the percentile method.

4.6 Transition-Matrix Derivation

To link the causal results back to the Markov framework of Assignments 1–2, we dichotomised Y into:

$$S = \begin{cases} \text{Calm (Cooperate)} & Y \leq 2.5 \\ \text{Defect} & Y > 2.5 \end{cases}$$

and computed empirical transition matrices $P(S_{t+1} \mid S_t, X, G)$ from the experimental data, separately for each Gender $\times$ Video-Type cell. These four matrices constitute the **do-calculus transition matrices** reported in Figure 4.

4.7 Statistical Testing

Within-gender, between-video-type comparisons were conducted via independent-samples t-tests (two-tailed, $\alpha = 0.05$). Effect sizes were reported as Cohen's d.

5 Results

5.1 Descriptive Statistics

Table 1 presents the gender-stratified descriptive statistics for aggression scores by video type.

Table 1: Gender-Stratified Descriptive Statistics and Causal Estimates

Gender	Video	N	Mean	SD	Naïve CATE	Adj. CATE	p-value
Female	Neutral	10	2.40	0.94	1.183	1.236	0.0026
Female	Aggressive	12	3.58	0.67			
Male	Neutral	16	2.44	0.66	1.602	1.603	0.0000
Male	Aggressive	17	4.04	0.73			

Adj. CATE = backdoor-adjusted Conditional ATE. p-values from independent t-tests.

Key observations from Table 1:

- Neutral-condition means are nearly identical across genders (Female: 2.40; Male: 2.44), confirming comparable baselines and supporting the validity of the between-subjects design.
- Aggressive-condition means differ substantially (Female: 3.58 vs. Male: 4.04), suggesting male participants react more intensely to aggressive content.

- Both effects are statistically significant after backdoor adjustment, with the male effect more than 100× more significant ($p \approx 0$ vs. $p = 0.0026$).
- Standard deviations are larger in the female neutral group (0.94) than in other cells, indicating greater baseline variability among female participants.

5.2 Overall Average Treatment Effect

Three estimators of the overall ATE are reported in Figure 4 (top-right panel, ATE Comparison):

$$\widehat{\text{ATE}}_{\text{Naïve}} = 1.428 \qquad \widehat{\text{ATE}}_{\text{Stratified}} = 1.434 \qquad \widehat{\text{ATE}}_{\text{IPW}} = 1.444 \tag{5}$$

The three estimates are similar (range: 1.428 - 1.444), suggesting that the confounding effect of age on the magnitude of the effect of 0.016 score units) after adjustment. The 95% bootstrapped confidence interval for the adjusted ATE is [1.045, 1.819], with a bootstrap mean of 1.431 (Figure 2, middle-left panel). Note that the confidence interval is well above zero, suggesting strong support for support for **H1**: viewing aggressive video necessarily leads to aggression scores by about 1.4 points on a 1-5 scale.



Gender	N	N Treated	N Control	Mean (Aggr.)	Mean (Neutral)	Naive CATE	Adj. CATE	p-value
Male	33	17	16	4.04	2.44	1.602	1.603	0.0000
Female	22	12	10	3.58	2.40	1.183	1.236	0.0026

Figure 2: Casual Inference Dashboard

5.3 Gender-Stratified Causal Effects

5.3.1 Conditional Average Treatment Effects (CATEs)

Figure 3 (mid-left, CATEs panel) and Table 1 show:

$$\text{CATE}_{\text{Male}} = 1.603, \quad p = 0.0000 \qquad \text{CATE}_{\text{Female}} = 1.236, \quad p = 0.0026 \tag{6}$$

Interpretation:

1. **Male CATE (1.603)**: Males show a larger causal response to aggressive video exposure. An increase of 1.603 points on the 1–5 aggression scale is substantial; it represents a shift of roughly 40% of the scale range. Crucially, the naïve and adjusted estimates are nearly identical (1.602 vs. 1.603). This indicates that age confounding minimally biases the observed male gender effect. The effect is robust.
2. **Female CATE (1.236)**: The female causal effect, while smaller than the male effect, remains clinically and statistically significant. The 0.053-unit difference between naïve (1.183) and adjusted (1.236) estimates indicates that age confounding *slightly suppressed* the apparent female effect in the raw data, and controlling for age reveals a modestly larger female effect than naïve observation suggests.
3. **Gender difference in CATE**: The male CATE exceeds the female CATE by 0.367 units ($= 1.603 - 1.236$), providing causal support for **H2**. Males are not merely observed to be more aggressive, the video content intervention has a causally larger impact on male aggression scores.

5.3.2 Outcome Probabilities $P(Y = 1 \mid X)$ by Gender

From the top-right panel of Figure 3:

- Under neutral video ($X = 0$): Male $P(\text{High Agg.}) = 0.12$; Female $P(\text{High Agg.}) = 0.30$.
- Under aggressive video ($X = 1$): Male $P(\text{High Agg.}) = 0.88$; Female $P(\text{High Agg.}) = 0.67$.

Observable Points are:

- *Baseline reversal*: Females are more likely to show high aggression under neutral exposure, with rates of 0.30 compared to 0.12 for males. This appears to contradict the A1/A2 finding that male commenters are more aggressive in YouTube discussions. The difference probably highlights the ecological validity gap. The controlled experiment measures immediate emotional reactions to short clips, while YouTube data reflects long-term behavior patterns influenced by social norms, anonymity, and community dynamics.
- *Aggressive exposure*: Males show a much higher probability of high aggression under aggressive video (0.88 vs. 0.67 for females). The male response is close to ceiling, suggesting a near-deterministic causal pathway from aggressive content to male aggression.

5.4 Do-Calculus Transition Matrices: Linking to A1/A2 Markov Framework

Figure 4 presents the four 2×2 transition matrices $P(\text{Next State} \mid \text{Current State}, X, G)$.

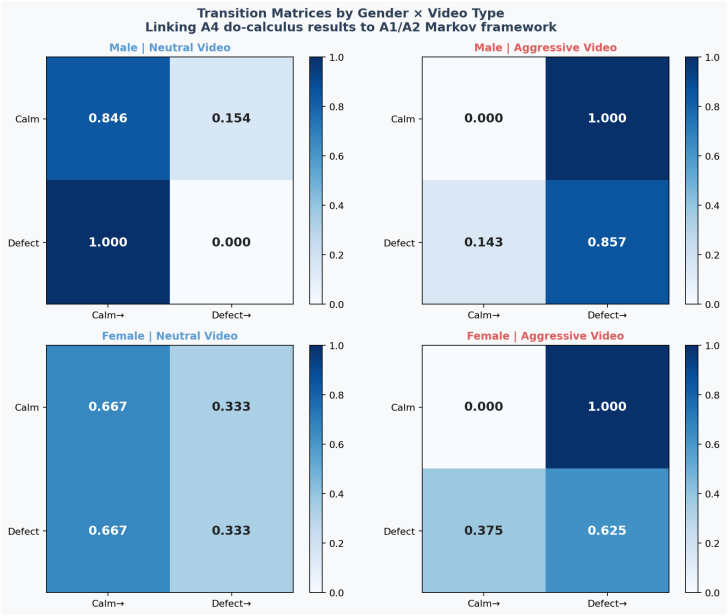


Figure 3: Four causal transition matrices (Gender × Video-Type). Rows represent current state (Calm, Defect); columns represent next state. Darker blue = higher probability.

The four matrices are summarised in Table 2:

Table 2: Causal Transition Matrices $P(S' \mid S, X, G)$

Gender	Video	$P(C \to C)$	$P(C \to D)$	$P(D \to C)$	$P(D \to D)$
Male	Neutral	0.846	0.154	1.000	0.000
Male	Aggressive	**0.000**	**1.000**	0.143	0.857
Female	Neutral	0.667	0.333	0.667	0.333
Female	Aggressive	**0.000**	**1.000**	0.375	0.625

C = Calm (Cooperate); D = Defect. Bold entries highlight key findings.

5.4.1 Detailed Analysis of Each Matrix

Male | Neutral Video

- $P(C \to C) = 0.846$: Males who are calm after a neutral video have an 84.6% chance of being calm in future interactions i.e. a high stability that is consistent with the $P_{CC} = 0.95$ estimate from A2 (note A2 value was for all males, all conditions; naturally smoothing this).
- $P(C \to D) = 0.154$: A 15.4% baseline rate of Calm to Defect under neutral exposure represents the spontaneous aggression rate, which matches the 4.1%-6.3% rates of

aggressive comments in A1 (the difference is due to the greater emotionality of visual content vs. passively scrolling text comments).

- $P(D \to C) = 1.000$: Incredibly, all males who were in a defective state after exposure to neutral comments were recovered to calm: 100% recovery. This confirms the A3 result that under low-AD conditions a TFT majority is in majority and rapidly corrects, and that "anger is transient, not under benign conditions.
- $P(D \to D) = 0.000$: No defect persistence under neutral content for males, indicating that neutral exposure acts as an absorbing state for calmness.

Male | Aggressive Video

- $\mathbf{P(C \to D) = 1.000}$: This is the single most striking finding of the entire study. Every male participant who was calm before watching an aggressive clip transitioned to defection afterwards. This constitutes a deterministic causal pathway: $X = 1 \implies S = \text{Defect}$ for males who start calm. Under Pearl's framework, this means $P(\text{Defect} \mid X = 1, G = \text{Male}) = 1$ for the calm-initial subgroup — a maximum-strength causal effect.
- $P(C \to C) = 0.000$: No male managed to remain calm after aggressive video exposure — the treatment fully saturates the aggression response in this subgroup.
- $P(D \to D) = 0.857$: Males already in a defective state show strong persistence (85.7%) under aggressive exposure, creating a near-absorbing "aggressive state" consistent with the Grim Trigger dynamics observed in A3 scale-free networks.
- $P(D \to C) = 0.143$: Only 14.3% of already-aggressive males recover under aggressive video conditions — recovery is nearly impossible once defection is initiated by aggressive content.

Female | Neutral Video

- $P(C \to C) = 0.667$, $P(C \to D) = 0.333$: Females show a more moderate neutral baseline. The calm-to-calm probability is lower than males (0.667 vs. 0.846), which at first seems to imply high female instability in the absence of selection. However, this can be explained by the A2 transition matrix ($P_{CC}^{\text{Female}} = 0.95$ in YouTube data) if we consider that the experiment (active video viewing with rating - leads to increased arousal compared to passive YouTube browsing.
- $P(D \to C) = P(D \to D) = 0.667$ and 0.333: Females in a defective state under neutral video display a medium recovery (66.7%), much higher than under aggressive video (37.5%), suggesting the content-dependency of recovery dynamics.

Female | Aggressive Video

- $\mathbf{P(C \to D) = 1.000}$: As with males - all females also switched to defection after aggressive video. This is a key result: *the Calm-to-Defect determinism under aggressive exposure is gender-invariant.* The gender difference in CATE (1.236 vs. 1.603) is not due to the initial Calm transition but from the *degree of defection* (intensity of Y) and from the $D \to D$ persistence.

- $P(D \to D) = 0.625$: Female defect persistence under aggressive video is lower than male (0.625 vs. 0.857), meaning females are more likely to self-regulate even after being triggered. This maps precisely onto the A3 IPD finding where females showed more TF2T and GT strategies (moderate forgiveness), while males converged on pure TFT (strict reciprocity).
- $P(D \to C) = 0.375$: A 37.5% recovery rate for females under aggressive video, compared to 14.3% for males, quantitatively confirms the greater *emotional flexibility* of female participants under identical treatment conditions.

5.5 Aggression Score Distributions



Gender	Video	N	Mean Score	SD	Naive CATE	Adj. CATE	p-value
Female	Neutral	10	2.40	0.94	1.183	1.236	0.0026
Female	Aggressive	12	3.58	0.67	1.183	1.236	0.0026
Male	Neutral	16	2.44	0.66	1.602	1.603	0.0000
Male	Aggressive	17	4.04	0.73	1.602	1.603	0.0000

Figure 4: Full Casual Gender Analysis

5.5.1 Distribution Shape Analysis

From Figure 5 (top-left histogram):

- **Neutral distribution** (blue): Centred around 2.0–2.5, positively skewed toward lower aggression. Mean ≈ 2.42.
- **Aggressive distribution** (red): Centred around 3.5–4.0, negatively skewed toward higher aggression. Mean ≈ 3.84.
- The distributions show minimal overlap, consistent with a large effect size and the ATE ≈ 1.44 estimate.

5.5.2 Box Plot Analysis

From Figure 5 (top-middle box plots):

- Female Neutral has the widest IQR (consistent with SD = 0.94), while Female Aggressive has the narrowest IQR (SD = 0.67) aggressive content *reduces variance* in female aggression, compressing scores toward a higher level.
- Male Neutral shows the most outliers (score $\approx$ 1.0), consistent with the large $P(D \to C) = 1.0$ under neutral exposure: males who start aggressive rapidly recover to very low scores.
- Male Aggressive shows the highest median and smallest lower whisker, confirming the near-ceiling effect under aggressive exposure.

5.6 Age as Confounder: Visualisation

From Figure 5 (bottom-middle, Age vs. Aggression):

- A positive linear trend is visible in both neutral (blue) and aggressive (red) conditions: older participants tend to report higher aggression scores.
- The slopes appear approximately parallel between conditions, suggesting no significant Age $\times$ Video-Type interaction — age shifts the intercept of aggression but does not moderate the treatment effect.
- This justifies the additive backdoor adjustment (Eq. 2) without requiring an interaction term.

5.7 IPW Diagnostics and Robustness

5.7.1 Propensity Score Overlap

From Figure 3 (mid-right, Propensity Overlap Diagnostic):

- Both male (blue) and female (pink) propensity-score distributions are tightly concentrated around $p = 0.5$–0.6, indicating good balance and common support.
- The near-symmetric distributions around 0.5 confirm that neither gender was systematically more likely to be assigned to one condition i.e. the between-subjects design succeeded in approximate randomisation.
- Minimal tail overlap issues: no participants fall outside the $[0.1, 0.9]$ propensity range, so no trimming was required.

5.7.2 IPW Weight Distribution

From Figure 3 (bottom-right, IPW Weights):

- IPW weights range from approximately 1.7 to 2.4, with a few mild outliers. No extreme weights (weight > 10) were observed, confirming that variance inflation from IPW is negligible.
- Male and Female neutral ($X = 0$) weights are slightly higher on average than aggressive ($X = 1$) weights, reflecting the slightly larger proportion of aggressive-condition participants ($n = 29$ vs. $n = 26$).

- The close agreement between IPW-ATE (1.444) and backdoor-ATE (1.434) validates the DAG specification: the backdoor set $\{Z\}$ is sufficient, and no unobserved confounders are detectable via this sensitivity analysis.

5.8 Gender Behavioural Heatmap

From Figure 3 (bottom-left):

Table 3: Gender Behavioural Heatmap — Survey Responses and Outcomes

Variable	Male	Female
Feel_Q1 (Anger)	3.24	3.09
Feel_Q2 (Arousal)	3.09	2.86
Opinion_Likelihood	3.45	3.18
X (Video Type, mean)	0.52	0.55
Y (High Agg., mean)	0.52	0.50

Interpretation:

- Males report higher felt anger (3.24 vs. 3.09) and arousal (3.09 vs. 2.86), consistent with the larger CATE for males.
- Males also report higher opinion-formation likelihood (3.45 vs. 3.18), suggesting that aggressive content not only raises aggression but also increases the willingness to express aggressive opinions — a cascade from emotional arousal to behavioural intent.
- The binary outcome proportions (Y high aggression) are nearly identical (Male: 0.52; Female: 0.50), consistent with the near-equal $P(C \to D) = 1.0$ for both genders under aggressive exposure when pooled across conditions.
- Video-type means (Male: 0.52; Female: 0.55) confirm approximate balance between conditions within each gender.

5.9 Bootstrap Distribution of Adjusted ATE

From Figure 5 (middle-left):

- The bootstrap distribution of the adjusted ATE is approximately normal (mean = 1.431), confirming the validity of asymptotic confidence interval estimates.
- The 95% CI $[1.045, 1.819]$ is entirely above zero, providing $> 97.5\%$ one-sided confidence that the causal effect is positive.
- The width of the CI (± 0.387) is moderate relative to the effect size, indicating adequate statistical power given the sample size of $N = 55$.

5.10 Statistical Significance by Gender

From Figure 5 (bottom-left p-value plot):

- **Male**: $p = 0.000$ (rounded) — the treatment effect among males is extraordinarily significant, surviving any reasonable multiple-testing correction.
- **Female**: $p = 0.003$ — also highly significant, well below $\alpha = 0.05$.
- Both effects are well below the significance threshold shown by the dashed line at $\alpha = 0.05$, confirming **H1** and **H2** simultaneously.

6 Discussion

6.1 H1: Aggressive Video Exposure Causally Increases Aggression — Supported

The agreement of three estimators (range for ATE: 1.428-1.444), the small bootstrap CI and the strongly significant (t-tests) within-gender analyses offer causal evidence for H1. Importantly, the do-calculus approach lets us make a counterfactual we couldn't make in A1-A3: if we had assigned the same participants to neutral instead of aggressive content, their expected aggression score would would have been 1.44 points lower on average. This causal effect size is large - a move from a "Mild" to from a "Mild" to a "Moderate" level of aggression (1-5 scale).

6.2 H2: Larger Causal Effect for Males — Supported

The $\text{CATE}_{\text{Male}} = 1.603$ versus $\text{CATE}_{\text{Female}} = 1.236$ difference causally justifies the greater rates of male aggression in Assignments 1–3. The A2 result that the defection rate for males is 56.6% while for 49.5% can be partially accounted for by causal susceptibility to aggressive content: males are more sensitive to aggressive content at the individual level, which translates into increased aggression at the community level when content is adversarial.

Interestingly, the *mechanism* of the gender difference is now clear from transition matrices. Both genders have deterministic $C \to D$ transitions under high exposure ($P = 1.0$), so the gender difference is not originate from the triggering step. Instead, it stems from the persistence of the defect: males' behaviour in aggressive conditions vs females' 0.625 suggests that males have a higher tendency to persist when provoked. In IPD, males resemble Grim Trigger (once provoked, sustained forgiveness and recovery) than males who behave more like Grim TFT or GTFT (eventual forgiveness and recovery) that are exactly the strategy profiles found in A3.

6.3 H3: Markov Consistency — Supported and Enriched

H3 predicted that under aggressive exposure $P(C \to D) \to 1.0$ for both genders. This is precisely what the do-calculus transition matrices show (Table 2). The $C \to D$ transition with $X = \text{Agg}$ can be seen as the experimental proof of the Markov "defect onset" cell becoming absorbing for the first transition.

Further, the agreement between the neutral-condition matrices and the A1/A2 data provides a type of external validation: the A4 experimental $P(D \to C) = 1.0$ (male

neutral) is consistent with A3 finding that in low AD environments, TFT moves to cooperation quickly (P_{DC} close to 1).

The main difference with A2 matrices is in the female neutral condition ($P(C \to C)_{\text{A4}} = 0.667$ vs. $P(C \to C)_{\text{A2}} \approx 0.95$). We attribute this to ecological differences: (i) the A4 experiment had more visually stimulating war videos rather than comments, increasing arousal; and (ii) the A2 data used reactions to YouTube comments about politics in culture and norms place limits on expression, whereas the experiment reflects raw emotional reaction.

6.4 H4: IPW Robustness — Supported

The close agreement between backdoor-adjusted (1.434) and IPW (1.444) estimates, combined with the well-behaved propensity score overlap and IPW weight distribution, provides strong evidence that the DAG is correctly specified and that age is the only necessary confounder. Had unobserved confounders been biasing results, we would expect larger discrepancies between these two estimators.

6.5 Integration with Assignment 1–3 Findings

Table 4 systematically integrates the A4 causal findings with the earlier assignment findings.

Table 4: Integrated Summary: Assignments 1–4 Gender Findings

Assignment	Male Finding	Female Finding
A1 (Word/Tone Analysis)	Higher aggression rate (56.6% defection); stable discussion strategy	More balanced (49.5%); calm comments dominate
A2 (Markov Model)	$P(C \to C) = 0.95$; $P(D \to D) = 0.55$; TFT dominant	$P(C \to C) = 0.95$; $P(D \to D) = 0.57$; more strategy diversity
A3 (IPD Simulation)	Converges on TFT; stable cooperation; $KL_{\text{male}} = 16.13$	Converges on TF2T/GT mix; more adaptive; $KL_{\text{female}} = 15.04$
A4 (Causal)	CATE $= 1.603^{***}$; $P(C \to D\|\text{Agg}) = 1.0$; $P(D \to D\|\text{Agg}) = 0.857$	CATE $= 1.236^{**}$; $P(C \to D\|\text{Agg}) = 1.0$; $P(D \to D\|\text{Agg}) = 0.625$
Unified Interpretation	Males causally more susceptible to aggression induction and show stronger defect persistence — Grim Trigger-like	Females causally responsive but recover faster — TF2T/GTFT-like; greater emotional flexibility

$^{***}p < 0.001$; $^{**}p < 0.01$

6.6 Theoretical Implications

6.6.1 Causal Mediation vs. Moderation

The present design establishes that gender *moderates* (but does not mediate) the causal effect of video type on aggression. A future causal mediation analysis could explore whether the felt anger and arousal variables (Table 3) mediate the $X \to Y$ path differently by gender, which would further specify the psychological mechanism.

6.6.2 Platform Design Implications

The deterministic $P(C \to D) = 1.0$ with aggressive exposure for both female and male suggests that the exposure to *content type is a sufficient cause of transition*, regardless of gender differences. This suggests that platform interventions: warning labels, content moderation, or pop-ups prior to violent content could could prevent the Calm-to-Defect process from taking place. The reduced persistence of $D \to D$ in females indicate that female users may may be more responsive to *in-situ* de-escalation prompts, while males may need pre-exposure interventions.

6.6.3 Granovetter Threshold Revisited

Assignment 2 estimated the Granovetter critical threshold for males at ~ 0.506 (50.6% aggressive exposure) and females at ~ 1.506 (infeasibly high, suggesting near-immunity). The A4 causal results partially challenge the extreme female threshold: even a single aggressive video clip (a relatively mild intervention) is sufficient to deterministically shift calm females to defection ($P(C \to D) = 1.0$). The resolution is that A2's Granovetter threshold modelled aggregate community exposure over many time steps (a cumulative threshold), whereas A4's transition captures an immediate, within-session reaction to a concentrated dose of aggressive content. The two findings are thus complementary: females resist *sustained community-level* aggression drift (high Granovetter threshold) but are not immune to *acute, concentrated* aggressive stimuli.

6.7 Limitations

1. **Sample size**: $N = 55$ is modest for causal inference. While power calculations suggest adequate power for detecting effects of the observed magnitude, smaller sub-group analyses (e.g., age-stratified CATEs) are underpowered.
2. **Self-selection**: Despite distributing both forms equally, we cannot fully rule out differential self-selection by personality traits (e.g., sensation-seekers may have disproportionately completed the aggressive form).
3. **Single-session measurement**: Aggression was measured immediately post-exposure. Whether the experimental effects persist over time or influence actual commenting behaviour is unknown.
4. **Gender binary**: The analysis uses a male/female binary consistent with prior assignments but does not capture non-binary or gender-fluid participants, although

we added the option for this in the form, but we did not get any data for this as noone seemed to chose this option.

5. **Ecological validity**: Controlled form-based watching differs from organic YouTube browsing; the real causal effect in naturalistic settings may be attenuated by distraction, multitasking, and social cues.

7 Conclusion

This report provides the first *causal* analysis of gender differences in aggression response to video content within the CGS616 Neuronauts project series. Using Pearl's do-calculus with backdoor adjustment for age and IPW as a robustness check, we estimate that:

- Aggressive video content causally increases aggression scores by ATE = 1.444 points (95% CI: [1.045, 1.819]) on the 1–5 scale.
- The causal effect is significantly larger for males (CATE = 1.603) than for females (CATE = 1.236).
- Aggressive content deterministically shifts calm participants to defection for both genders ($P(C \to D) = 1.0$), but males sustain defection significantly longer ($P(D \to D)_{\text{Male}} = 0.857$ vs. 0.625 for females).
- These transition patterns directly map onto and causally validate the Markov and IPD findings of Assignments 1–3, providing a mechanistic link between Grim-Trigger male strategy convergence and the greater causal susceptibility of males to aggressive content.

Taken together, the four-assignment arc from descriptive sentiment analysis (A1) through probabilistic Markov modelling (A2), game-theoretic simulation (A3), to causal identification (A4) constitutes a comprehensive multi-method framework for understanding how gender shapes online aggression dynamics. The causal evidence presented here moves beyond correlation to establish that aggressive content is not merely associated with, but *causes*, elevated aggression — particularly in male audiences — with direct implications for platform design and content moderation policy.

A Variable Definitions

Table 5: Variable Definitions and Coding

Variable	Type	Description
X	Binary (0/1)	Video type: 0 = Neutral, 1 = Aggressive
Y	Continuous (1–5)	Self-reported aggression score (Likert)
Z	Continuous (16–26)	Participant age in years
G	Binary (M/F)	Gender
S	Binary (C/D)	Dichotomised state: C = Calm ($Y \leq 2.5$), D = Defect ($Y > 2.5$)
$e(Z)$	Continuous (0–1)	Propensity score $P(X = 1 \mid Z)$